This CVPR Workshop paper is the Open Access version, provided by the Computer Vision Foundation. Except for this watermark, it is identical to the accepted version; the final published version of the proceedings is available on IEEE Xplore.

Physics-Based 3D Ball Trajectory Reconstruction from Monocular Soccer Video: A Multi-Model Benchmark

Łukasz Grad*
University of Warsaw, Poland
ReSpo.Vision, Poland
l.grad@mimuw.edu.pl

Krzysztof M. Czajkowski*
University of Warsaw, Poland
ReSpo.Vision, Poland
k.czajkowski@uw.edu.pl

Aliaksandr Varashylau*
SplitMetrics, Poland
waraszylau@gmail.com

Abstract

We study physics-based 3D ball trajectory reconstruction from monocular videos. Our pipeline segments the trajectory at contact events and fits each segment by optimizing a forward-simulated flight model with a reprojection-based objective. We implement seven arc models of increasing complexity, ranging from gravity-only parabolas to MuJoCo simulations with drag, spin, and fluid forces. All seven are evaluated under end-to-end monocular reconstruction, with four additionally evaluated under oracle 3D fitting. The two protocols yield reversed model rankings: a model with fitted gravitational acceleration achieves the best monocular accuracy on all soccer evaluation settings, yet a spin-decomposition model leads on four of five datasets when fitted to 3D ground truth. These results suggest that observation noise and single-view geometric ambiguity, rather than model expressiveness, are the primary limiting factors for monocular soccer reconstruction. We evaluate across five datasets spanning nearly 6,000 trajectory segments and publicly release two new soccer datasets with triangulated 3D ground truth along with segment-level annotations for APIDIS and ISSIA-3D. Project page and code: lukaszgrad.github.io/soccer-ball3d.

1. Introduction

Knowing the 3D trajectory of the ball is essential for analytics tasks that go beyond 2D tracking: measuring shot elevation, pass apex height, or free-kick curvature requires the ball's position in three dimensions, as do broadcast overlays that must respect ball height. Current commercial systems reconstruct 3D positions from dedicated multi-camera arrays, but such installations are expensive and unavailable at multiple competition levels, while fixed tactical cameras and broadcast feeds, from which a single-view reconstruction could operate, are already widely deployed. Since depth is fundamentally ambiguous from single-view, we use physics-based flight models to constrain the space of plausible 3D trajectories.

In this paper, we address the inverse problem of recovering the ball's 3D flight from per-frame 2D detections, camera calibration, and contact-event timestamps (pivot points). Ball detections and camera calibration are obtained automatically; pivot points are manually annotated. We focus on the trajectory reconstruction problem itself. The core challenge is fitting a physical model to noisy image-plane observations: the optimizer must recover initial position, velocity, and aerodynamic parameters from 2D reprojection residuals alone, where distinct spin states can produce nearly identical projections. A natural question arises: does increasing model complexity improve reconstruction, or do observation noise and single-view ambiguity render the extra parameters unrecoverable?

Prior monocular methods each commit to a single flight model and do not compare protocols that isolate observation quality from model capacity, leaving this trade-off unexplored. Our main contributions are as follows:

- **Reconstruction pipeline.** A monocular 3D trajectory fitting pipeline that segments the flight at pivot points, selects between straight and arc models, refines boundaries near high pivots using player foot-points, and rejects unreliable segments (Section 4).
- **Physics model benchmark.** Seven arc models spanning ODE and MuJoCo formulations, with progressively added drag, spin, and fluid forces (Section 3).
- **Datasets and benchmarks.** Two publicly released soccer datasets with triangulated 3D ground truth (Legia-Piast, Stalowa Wola), a broadcast benchmark with aggregate results (Ekstraklasa-Broadcast), and segment-level annotations for APIDIS and ISSIA-3D (Section 5).
- **Experimental results.** All seven models evaluated under end-to-end monocular reconstruction, with four also under oracle 3D fitting. We propose a detection-style mAP metric for 3D ball trajectory evaluation that jointly

*The authors contributed equally to this research.

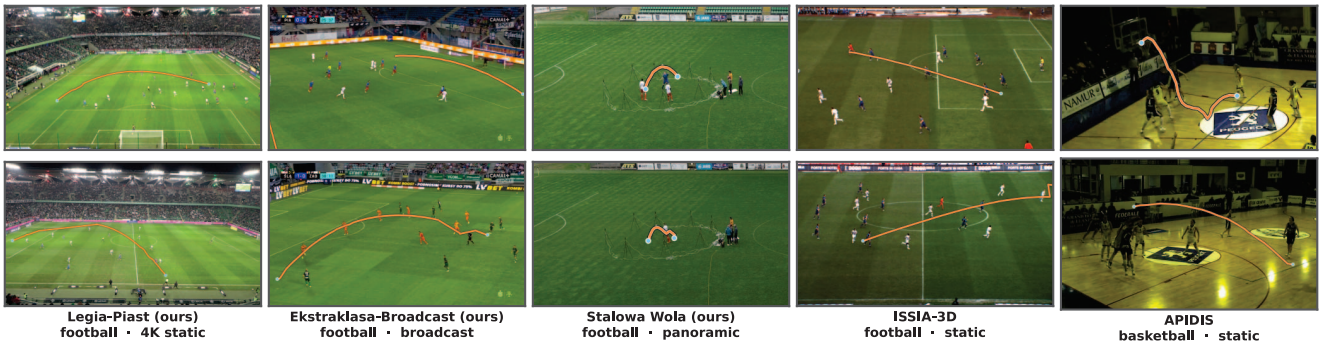


Figure 1. **Dataset overview.** Each column shows one of the five evaluation datasets from two different viewpoints or sequences. Orange curves depict ground-truth 3D ball trajectories projected onto the image; light-blue dots mark segment endpoints (pivot points). The datasets span soccer and basketball, static multi-camera and dynamic broadcast setups, and resolutions from 1080p to 4K (Table 2).

captures reconstruction error and segment coverage (Section 6).

2. Related work

Physics-based monocular 3D trajectory reconstruction. Fitting a physical flight model to 2D ball observations from a calibrated monocular camera has been explored across multiple sports. In soccer, early work reconstructed 3D scenes from stereo video via geometric splines [1], while Yamada *et al*. [24] fit a parabolic model after per-frame calibration from field lines. Metzler and Pagel [20] extended this to static-camera setups with motion-history-based detection and physics-constrained 3D fitting, reporting mismatches when real trajectories deviated from ideal parabolas. In basketball and volleyball, Chen *et al*. [5] and Huang *et al*. [11] pioneered monocular reconstruction with gravity-only projectile models, using court geometry for calibration. More recent work has incorporated aerodynamic forces. MonoTrack [17] builds a full monocular pipeline for badminton with a quadratic-drag model and GRU-based shot segmentation. In table tennis, Calandre *et al*. [4] extract kinematic parameters including drag and Magnus coefficients from a single camera via ball-diameter-based depth estimation. TT3D [9] fits a drag-and-Magnus ODE via 2D reprojection on synthetic data, and LATTE-MV [8] processes 73,000 exchanges with a Stokes-drag model for robot anticipation. Despite this progression, each prior reconstruction system commits to a *single* flight model, so reported accuracy conflates the flight model with the rest of the pipeline. None evaluates under oracle 3D inputs, leaving open whether observation quality or model complexity governs ranking: the question our seven-model, five-dataset, dual-protocol comparison answers.

Learning-based 3D trajectory estimation. An alternative to per-segment physics optimization is to learn a mapping from 2D observations to 3D trajectories, typically by training on simulated data. Ponglertnapakorn and Suwajanakorn [22] propose an LSTM pipeline with a camera-independent canonical representation, trained entirely on synthetic trajectories and evaluated across tennis and soccer. SynthNet [7] similarly predicts initial conditions with an MLP under a drag-only model, though [22] reports substantially higher landing errors than optimization-based alternatives. Kienzle *et al*. [14] train a Spin Prediction Transformer on MuJoCo-generated table tennis trajectories to jointly predict 3D position and spin from 2D keypoints, while their earlier work [13] uses physics equations as a training-time loss for weakly-supervised 3D localization. In golf, McNally *et al*. [19] embed a neural network inside a projectile-motion ODE to predict aerodynamic coefficients from accurate launch-monitor data, a forward-simulation setting with precise 3D inputs rather than the inverse problem from noisy 2D. Kim *et al*. [15] sidestep vision entirely, inferring 2D ball position from player tracking data via a Set Transformer. These methods amortize the inverse problem but require sport-specific training data and generally do not expose individually interpretable physical parameters (with partial exceptions such as [14, 19]). Our optimization-based fitting stage requires no training data but assumes a calibrated camera model, pivot timestamps, tracked 2D ball detections, and player detections for contact-point correction.

Per-frame 3D ball localization and sports datasets. A complementary line of work estimates ball depth from single calibrated images without temporal modeling. Van Zandycke and De Vleeschouwer [23] predict the ball's apparent pixel diameter via a CNN and recover 3D position from the known physical size, while Caio *et al*. [3] improve on this by predicting the ball's vertical ground-plane projection using contextual scene cues. SoccerNet-v3D [10] constructs 3D ball annotations for soccer via multi-view triangulation and proposes a ball-size-prior monocular baseline,

but reports 6–14 m errors for just 10% bounding-box size variation, quantitatively motivating the need for physics-based temporal constraints. Among existing sports datasets, SoccerNet-v3D [10] provides per-frame 3D ball positions for soccer from temporally disconnected replay pairs but no trajectory-segment annotations; APIDIS [6] offers a single three-minute indoor multi-camera basketball sequence; and DeepSportradar-v2 [12] defines a per-frame ball 3D localization task. None of these benchmarks provides dense 3D ball ground truth *with* trajectory-segment annotations at the scale needed for systematic model comparison. We address this gap with two new soccer datasets with triangulated 3D ground truth, a broadcast benchmark with aggregate results, and segment-level annotations for two existing datasets, forming a five-dataset benchmark with nearly 6,000 segments.

Prior work reports diverse metrics: mean absolute 3D error [9], 2D reprojection error [17], normalized RMSE [22], and spin classification accuracy [14]. For the monocular protocol we introduce a detection-style mAP that jointly accounts for reconstruction error and segment coverage in a single metric; for the oracle 3D protocol we report frame-weighted mean 3D error.

3. Physics models

We employ two classes of models for the segment fitting procedure of Section 4.3: a straight on-ground model for surface-level movement and several arc models of increasing complexity for airborne trajectories. Table 1 summarizes the arc models and their free parameter counts.

3.1. Straight on-ground motion model

The straight motion model assumes a constant deceleration of magnitude k along the direction of motion in the x-y plane. Given known initial and final positions, the trajectory follows:

$$\begin{aligned} x(t) &= x_{\text{start}} + (v_0 t - \tfrac{k}{2}t^2)\cos\theta \\ y(t) &= y_{\text{start}} + (v_0 t - \tfrac{k}{2}t^2)\sin\theta \\ z(t) &= 0 \end{aligned} \tag{1}$$

with $v_0 = \frac{D + \frac{k}{2}T^2}{T}$, where D is the displacement magnitude, T the trajectory duration, and θ the direction angle in the x-y plane.

3.2. Arc motion models

The simplest arc model, *basic parabola*, sets $k_3 = k_l = k_s = 0$, yielding a pure ballistic trajectory under gravity alone with no free physics parameters beyond the initial state. We consider two airborne model families: the formulation outlined in [21] and the ellipsoid model implemented in the MuJoCo simulation engine.

The arc models based on the formulation of [21] describe the motion using Newton's second law with the force comprised of gravity, aerodynamic drag opposing forward motion, and the Magnus force, which acts perpendicular to linear and angular velocity of the ball. In this formulation the Magnus force is decomposed into two components: lift, acting along the unit vector $\boldsymbol{\ell}$, and lateral force, acting along $\boldsymbol{\ell} \times \mathbf{v}$, where $\boldsymbol{\ell}$ is defined as the direction perpendicular to the velocity $\mathbf{v}$ that lies in the $\mathbf{v}$-z plane. The general equations of motion, with air resistance proportional to $|v|^2$, are:

$$\begin{aligned} x'' &= -k_3|v|x' + k_l|v|\frac{x'z'}{|v_p|} + k_s|v|^2\frac{y'}{|v_p|} \\ y'' &= -k_3|v|y' + k_l|v|\frac{y'z'}{|v_p|} - k_s|v|^2\frac{x'}{|v_p|} \\ z'' &= -g - k_3|v|z' - k_l|v||v_p| \end{aligned} \tag{2}$$

where primes denote time derivatives, k_3 is the drag coefficient, k_l the lift coefficient, k_s the side-spin Magnus coefficient, $|v|$ the velocity magnitude, $|v_p|$ the magnitude of its x-y projection, and g gravity.

The *basic kinetic* model solves Equation 2 with $k_l = k_s = 0$, retaining only the drag coefficient k_3 as a free aerodynamic parameter. The *basic fitg* variant additionally optimizes the gravitational constant g, which acts as a cheap proxy for unmodelled forces. The *basic angular* model activates both k_l and k_s, decomposing spin effects into lift (topspin/backspin) and lateral (sidespin) components without tracking angular velocity explicitly. All three variants are integrated with an adaptive-step ODE solver.

3.3. MuJoCo arc motion model

As an alternative to the ODE models above, we employ MuJoCo, a rigid-body physics engine with built-in fluid-force models that account for drag, viscous resistance, and torque acting on the ball. In its simplest configuration (*MuJoCo kinetic*), fluid-force coefficients are fixed rather than fitted, and the ball experiences gravity and drag from MuJoCo's inertia-based fluid model with parameters derived from fluid density and body inertia. The *MuJoCo angular* variant additionally optimizes initial angular velocity $\boldsymbol{\omega}_0$. The optimized spin affects motion only through MuJoCo's fixed inertia-based fluid terms, and no explicit Magnus or lift coefficient is fitted.

MuJoCo provides two fluid-force models: an inertia-based model that computes force and torque from quadratic drag, and an ellipsoid model with five fluid coefficients for blunt-body drag, slender-body drag, angular drag, Kutta lift, and Magnus lift. For a spherical ball, Kutta lift vanishes, slender-body drag becomes negligible, and added mass is evaluated analytically, reducing the fitted coefficients to three: C_D^{blunt}, C_D^{ang}, and C_M. The full force decomposition is given in the supplementary material (Section 8.4).

Table 1. Comparison of arc physics models for projectile motion simulation. All models share the same straight (on-ground) model with 5 parameters (k, start/end positions). N_θ denotes the number of free parameters optimized per arc segment. All models share 6 initial-condition parameters ($\mathbf{r}_0, \mathbf{v}_0$) and differ in additional physics coefficients. * Gravity is fitted rather than fixed, acting as a proxy for unmodelled forces. † MuJoCo's inertia-based fluid model provides drag with fixed parameters derived from fluid density and body inertia.

Model	Framework	Gravity	Drag	Magnus	Spin ($\boldsymbol{\omega}_0$)	N_θ
Basic parabola	ODE	fixed	–	–	–	6
Basic kinetic	ODE	fixed	k_3	–	–	7
Basic fitg	ODE	fitted*	k_3	–	–	8
Basic angular	ODE	fixed	k_3	k_l, k_s	–	9
MuJoCo kinetic	MuJoCo	fixed	fixed†	–	–	6
MuJoCo angular	MuJoCo	fixed	fixed†	–	✓	9
MuJoCo ellipsoid	MuJoCo	fixed	$C_D^{\text{blunt}}, C_D^{\text{ang}}$	C_M	✓	12

4. Method

4.1. Input assumptions and pipeline overview

We address the problem of reconstructing 3D ball trajectories from monocular video. The method requires four inputs per sequence: (i) a per-frame camera model (intrinsics and extrinsics), possibly corrupted by calibration noise, (ii) pivot-point timestamps delimiting trajectory segments, (iii) tracked and smoothed 2D ball bounding boxes, from which a per-frame apparent ball size s_i (bounding-box diagonal) is derived, and (iv) player detections with foot-point proxies for correcting contact positions at high-contact pivots.

Given these inputs, the pipeline proceeds in three stages. First, the trajectory is segmented at pivot timestamps (Section 4.2). Second, each segment is independently fitted by minimizing the 2D reprojection residual between a forward-simulated 3D trajectory and the observed detections (Section 4.3). The physics model is treated as a black-box mapping $f_{\boldsymbol{\theta}}(t) \to (x, y, z)$. Specific instantiations are described in Section 3. Third, contact positions at high-contact pivots are corrected (the ball's vertical image coordinate is replaced by the nearest player's foot-point) and the adjacent segments are re-fitted with the updated boundary. Segments whose fitting residual exceeds a threshold are then rejected. The accepted segments are concatenated to form the final 3D trajectory.

4.2. Trajectory segmentation

Pivot timestamps partition the trajectory into contiguous motion phases (e.g. pass, bounce, header). All ball-ground and ball-player contact timestamps are labeled for soccer data. Pivot timestamps therefore correspond to events marking the beginning and end of arc trajectories such as passes and shots. This labeling scheme slightly limits the ball trajectory coverage, but keeps the trajectories most valuable for benchmarking the physics models (see Appendix 8.3 for coverage statistics). Each pivot is classified as either a *ground contact* (low pivot) or a *high-contact pivot* (e.g. header, chest control). The classification is rule-based: a pivot is labeled as high-contact if (a) a nearby player's bounding box places the ball above a height threshold and the adjacent segments are not both straight, or (b) the ball's pitch-plane projection lies outside the field boundary while a player is nearby.

For high-contact pivots, projecting a head-height ball onto the ground plane produces a spurious contact location. The ball's vertical image coordinate is therefore replaced by the nearest player's foot-point coordinate, and the corrected position is back-projected onto the pitch plane to obtain updated pitch coordinates for the two adjacent segments.

4.3. Segment fitting

Each segment is fitted independently using a two-stage cascade: a straight on-ground model is fitted first by nonlinear least squares. If its residual exceeds a threshold, an arc model is also fitted, and the variant with the lower residual is selected. Segments whose length falls outside a valid range or whose best-fit residual exceeds an acceptance threshold are rejected.

The physics model produces a 3D trajectory $(x(t), y(t), z(t))$ for a given parameter vector $\boldsymbol{\theta}$, which is projected into the image via the per-frame camera model. The optimizer adjusts $\boldsymbol{\theta}$ to minimize the composite residual defined in Section 4.4. Straight segments are solved with Levenberg–Marquardt [16, 18]. Arc segments use trust-region reflective least squares with box constraints [2].

Refinement. After the first-pass fit, each pivot relabeled as high-contact receives the corrected contact position described in Section 4.2. Only the two adjacent segments are re-fitted. All other segments are retained from the first pass. This local second pass enforces spatially consistent boundary conditions at airborne ball–player interactions without re-optimizing the full sequence. Edge cases are handled conservatively: segments that are too short or too long are

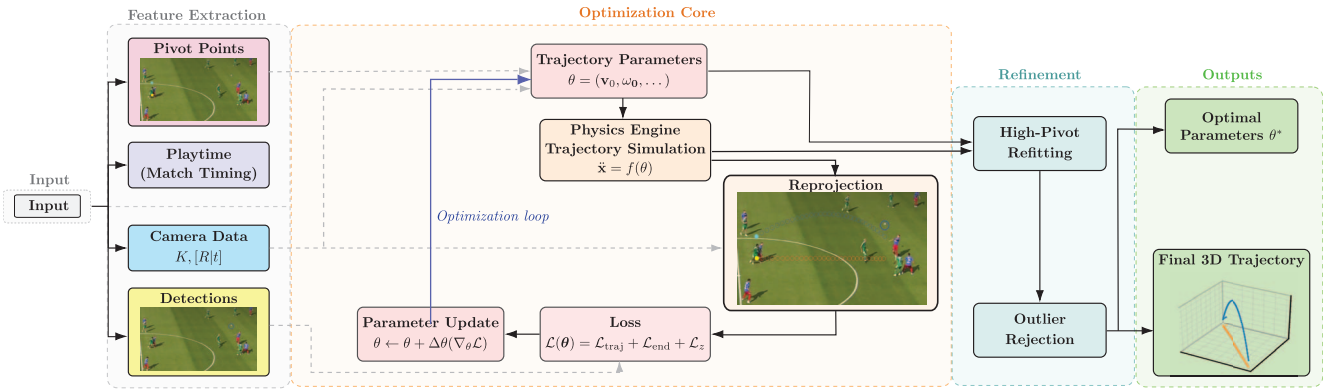


Figure 2. **Overview of the 3D Trajectory Reconstruction Pipeline.** A monocular input video, ball and player features and camera calibration $(K, [R|t])$ are given. Pivot points and playtime from manual annotation segment the sequence, which is fitted in an optimization loop: a physics engine simulates and reprojects the 3D path to iteratively improve trajectory parameters. Finally, the pipeline corrects high-contact pivots, re-fits the affected segments, and rejects invalid segments.

skipped, and those whose best-fit residual still exceeds the acceptance threshold are rejected. For shot-like segments with unreliable endpoints, only the valid prefix is re-fitted without the endpoint constraint.

4.4. Loss function

The fitting objective is a residual vector rather than a scalar loss, enabling Levenberg–Marquardt and trust-region solvers to exploit the least-squares Jacobian structure. It comprises three terms.

Trajectory reprojection residual. For each frame i with a valid detection,

$$\mathbf{d}_i = \left|\mathbf{p}_{\text{model}}(t_i) - \mathbf{p}_{\text{obs}}(t_i)\right|, \tag{3}$$

where $\mathbf{p}_{\text{model}}(t_i) \in \mathbb{R}^2$ is the image-plane projection of the simulated 3D position and $\mathbf{p}_{\text{obs}}(t_i) \in \mathbb{R}^2$ is the observed ball center. Each component is normalized by the apparent ball size s_i:

$$\tilde{\mathbf{d}}_i = \frac{\mathbf{d}_i}{s_i}, \tag{4}$$

making the residual approximately scale-invariant to camera distance.

Height penalty (arc segments only). A piecewise penalty on the final simulated height z_T penalizes implausible endpoints:

$$\ell_z = \max\!\left(\tfrac{1}{2}\, z_T,\ c \cdot \left(|z_T - z_{\max}| - z_{\max}\right)\right), \tag{5}$$

where $z_{\max}$ is a height threshold and c is a scaling constant. For $0 \le z_T \le 2z_{\max}$ the penalty reduces to $\frac{1}{2} z_T$. Outside this range it grows linearly with slope c. Default values are $z_{\max} = 1.2\,\text{m}$ for segments with a known endpoint and $z_{\max} = 5\,\text{m}$ for open-ended segments such as shots (the penalty remains mild up to $2.4\,\text{m}$ and $10\,\text{m}$, respectively).

Weighted objective. The optimizer minimizes

$$\mathcal{L}(\boldsymbol{\theta}) = \mathcal{L}_{\text{traj}} + \mathcal{L}_{\text{end}} + \mathcal{L}_z\,, \tag{6}$$

where

$$\mathcal{L}_{\text{traj}} = \sum_{i=1}^{N} \left\|\tilde{\mathbf{d}}_i\right\|^2, \tag{7}$$

$$\mathcal{L}_{\text{end}} = N_s \left\|\tilde{\mathbf{d}}_1\right\|^2 + N_e \left\|\tilde{\mathbf{d}}_N\right\|^2, \tag{8}$$

$$\mathcal{L}_z = N_z\, \ell_z^2\,, \tag{9}$$

and $N_s = \lfloor w_s N + \frac{1}{2} \rfloor$, $N_e = \lfloor w_e N + \frac{1}{2} \rfloor$, $N_z = \lfloor w_z N + \frac{1}{2} \rfloor$ are integer weights proportional to the segment length N.

The elevated endpoint weights compensate for unreliable detections near pivot points. For *straight* segments, both start and end weights are active ($w_s > 0$ and $w_e > 0$). For *arc* segments, the start penalty is omitted ($w_s = 0$), and the end penalty uses the 2D pitch-plane displacement between the simulated endpoint and the known pivot position, normalized by the median ball size. In practice, the weighting is implemented by replicating residual entries, avoiding a custom weight matrix.

Appendix 8.1 reports an ablation study of these three terms on LP-static. The test results show that each of the two auxiliary terms helps stabilize the prediction, and removing both clearly degrades performance.

5. Data

Open-source sports datasets with per-frame 3D ball ground truth and segment-level annotations (pivot points, playtime) are scarce, limiting trajectory-level benchmarking of physics-based estimation methods. We publicly release two new soccer datasets with triangulated 3D ground truth, report aggregate results on a broadcast benchmark, and contribute segment-level annotations for two existing public datasets (see Table 2 and Figure 1).

Dataset	Sport	Cameras	Resolution	FPS	Dur. (min)	Sequences	Segments	Frames	View-frames	Play (%)	Arc (%)	Release
LP (ours)	soccer	6s + 1d	4K / 1080p	25	94	2	2 818	141k	845k	55.8	24.8	Full
SW (ours)	soccer	9s†	1080p / pano	30 / 25	8.5	4	518	13k‡	13k‡	73.5	33§	Full
EB (ours)	soccer	1d	1080p / 720p	25	50	5	2 384	74k	74k	56.0	26.6	Results
APIDIS [6]	basketball	7s	1600×1200	∼22	3	1	55	3.7k	15k	7.1	100	Annotations
ISSIA-3D [10]	soccer	6s	1080p	25	2	1	113	3.0k	18k	65.1	32.4	Annotations

Table 2. Overview of evaluation datasets. s = static camera, d = dynamic (broadcast). View-frames: raw frame count across evaluation views. Play(%): fraction of frames outside manually annotated pauses; Arc(%): fraction of active-play frames classified as airborne from 3D ground truth (see Appendix 8.2 for details). Per-frame split labels released for all datasets except EB (licensing restrictions). Release: Full = all assets, Results = aggregate metrics only, Annotations = our segment annotations for an existing public dataset. †Eight static cameras for triangulation and one stitched panoramic camera for evaluation. ‡Panoramic view only. §Available for two of four sequences.

Legia-Piast and Stalowa Wola. The *Legia-Piast* (LP) dataset captures a full Polish Ekstraklasa match (December 2024) recorded with six static 4K cameras (3840×2160 at 25 fps) and one tactical broadcast camera (1920×1080 at 25 fps), totaling 94 minutes (two halves). Per-frame 3D ball ground truth is obtained via multi-camera triangulation from the six static views. We evaluate on the broadcast camera and five static cameras, yielding 12 view-sequences. Released assets include videos, camera calibrations, 3D ground truth, ball detections, pivot-point annotations, and playtime labels.

The *Stalowa Wola* (SW) dataset comprises four short training-drill sequences (1.4–3.2 minutes each, 8.5 minutes total) recorded on a smaller pitch with eight static cameras (1920×1080 at 30 fps) used exclusively for triangulation. A separate stitched panoramic camera (7500×1200 at 25 fps) serves as the single evaluation view. Two to three players perform realistic passing and shooting drills. All videos, calibration data, 3D ground truth, and annotations are publicly released.

Ekstraklasa-Broadcast. The *Ekstraklasa-Broadcast* (EB) dataset consists of five 10-minute broadcast clips from Polish Ekstraklasa matches (2023–24 season) recorded at 25 fps with a single dynamic broadcast camera per match (four at 1920×1080, one at 1280×720). Per-frame camera calibration is provided. Ground-truth 3D ball positions are obtained from a proprietary multi-camera tracking system. The triangulation procedure is not disclosed. Due to licensing restrictions, neither the video footage nor the raw ground truth can be publicly released. We report aggregate evaluation results only.

APIDIS and ISSIA-3D. We adapt two public benchmarks. The *APIDIS* basketball dataset [6] is a three-minute indoor game sequence with seven synchronized static cameras. We evaluate on four. We use APIDIS as it is an established benchmark for 3D ball trajectory estimation that also lets us test whether the framework generalizes beyond soccer. The *ISSIA-3D* soccer dataset [10] is a two-minute outdoor match segment with six synchronized static cameras. For both datasets, we contribute manually annotated pivot points and playtime labels, enabling trajectory-level segmentation and benchmarking that the original releases lack. These annotations are publicly released.

3D ground truth generation. For LP and SW, 3D ball ground truth is computed via multi-ray triangulation. In each frame, ball detections from $N \geq 2$ cameras are back-projected into rays using pre-calibrated camera models, and the 3D point minimizing the sum of squared ray-point distances is estimated with RANSAC-based outlier rejection. The resulting ground truth achieves a median 3D ray-point distance of approximately $0.2\,\mathrm{m}$ on LP and $0.1\,\mathrm{m}$ on SW, computed across all camera rays and valid frames. Ground-truth quality is further validated through the 3D fitting experiments presented in Section 6, where the highest-capacity physics models achieve lower fitting error on the 3D annotations than simpler ones.

6. Results

6.1. Evaluation protocols and metrics

We evaluate trajectory reconstruction under two complementary protocols. Within each protocol, the compared models share all non-physics inputs and segment acceptance rules, so only the arc physics varies. All models use the same straight on-ground model.

End-to-end monocular reconstruction (2D fitting). The full pipeline (Section 4) processes 2D ball detections from a single camera view and produces per-segment 3D trajectory estimates. We adopt mean average precision (mAP) as the primary accuracy metric, adapting the AP formulation from object detection. Each predicted frame is treated as a detection whose confidence equals the fitting RMSE

Table 3. End-to-end monocular reconstruction results. Balanced mAP (mAP_{bal}) and arc-specific mAP (mAP_{arc}). Higher is better. APIDIS contains only arc segments, so $mAP_{bal} = mAP_{arc}$. Best per column in **bold**. Abbreviations and dataset details as in Table 2.

	LP-static		LP-broadcast		EB		SW		ISSIA-3D		APIDIS	
Model	mAP_{bal}	mAP_{arc}	mAP_{bal}	mAP_{arc}	mAP_{bal}	mAP_{arc}	mAP_{bal}	mAP_{arc}	mAP_{bal}	mAP_{arc}	mAP_{bal}	mAP_{arc}
basic parabola	.325	.274	.493	.330	.533	.357	.421	.351	.260	.159	**.577**	**.577**
basic kinetic	.332	.289	.508	.359	.532	.359	.414	.355	.253	.153	.476	.476
basic fitg	**.343**	**.307**	**.530**	**.397**	**.557**	**.412**	**.459**	**.371**	**.262**	**.163**	.507	.507
basic angular	.336	.296	.517	.377	.544	.383	.412	.353	.255	.157	.463	.463
MuJoCo kinetic	.329	.282	.496	.337	.525	.341	.422	.352	.260	.157	.552	.552
MuJoCo angular	.330	.283	.499	.341	.527	.345	.422	.351	.260	.158	.543	.543
MuJoCo ellipsoid	.324	.271	.494	.334	.522	.339	.437	.367	.261	.160	.547	.547

Table 4. Oracle 3D fitting results. Frame-weighted mean 3D error $\bar{e}$ (dm) and arc-specific error $\bar{e}_a$ (dm). Lower is better. Per-frame errors capped at 25 dm. Best per column in **bold**. Abbreviations as in Table 2.

	LP		EB		SW		ISSIA-3D		APIDIS	
	$\bar{e}$	$\bar{e}_a$	$\bar{e}$	$\bar{e}_a$	$\bar{e}$	$\bar{e}_a$	$\bar{e}$	$\bar{e}_a$	$\bar{e}$	$\bar{e}_a$
parabola	3.16	4.68	3.80	5.88	1.92	7.24	5.90	3.64	4.55	4.55
kinetic	3.20	4.74	3.68	5.26	1.86	6.05	5.54	2.80	3.68	3.68
fitg	2.75	3.12	3.48	4.14	**1.78**	5.89	5.34	2.36	3.12	3.12
angular	**2.24**	**1.83**	**3.28**	**3.32**	1.83	**5.77**	**5.22**	**2.08**	**2.56**	**2.56**

of its parent segment (lower is better). All frames within a segment share one confidence score. At each 3D distance threshold $T \in \{0.25, 0.5, 0.75, 1.0, 1.5, 2.0, 2.5, 3.0\}$ m, a detection is a true positive if the 3D Euclidean error is below T and a false positive otherwise. Ground-truth frames without predictions are false negatives. AP at each threshold is computed via VOC2010+ interpolation (monotone-decreasing precision envelope), and mAP is the mean of the eight per-threshold AP values. The primary metric is the balanced $mAP_{bal} = (mAP_{str} + mAP_{arc})/2$, which gives equal weight to both motion types. When only one split has ground-truth frames, mAP_{bal} equals that split's mAP. No train/test split is applied, as each segment is fit independently without learned cross-segment parameters. The evaluation set for each view comprises all non-pause frames with valid 3D ground truth. No action-specific or scenario-based subsetting is applied. The arc/straight split is derived from 3D ground truth (Appendix 8.2).

Oracle 3D fitting. The 3D fitting protocol bypasses detection and camera calibration by fitting models directly to 3D ground truth positions. This isolates model capacity from observation noise. We report the playtime frame-weighted mean 3D error ($\bar{e}$, dm), with per-frame errors capped at 25 dm to limit the influence of catastrophic fits. We evaluate four models that span the full ODE complexity hierarchy: *basic parabola* (gravity only), *basic kinetic* (drag), *basic fitg* (free gravity), and *basic angular* (spin decomposition).

Aggregation. All reported values are macro-averages: we first compute the metric independently for each evaluation unit (camera-half, match, or sequence), then average across units within a dataset. For SW, arc-specific metrics average over the two sequences that contain arcs (sequences 5 and 6). Sequences 39 and 42 are all-straight.

6.2. End-to-end monocular reconstruction

Table 3 summarizes the 2D fitting results across six evaluation settings. The *basic fitg* model achieves the highest mAP_{bal} on all five soccer settings, while *basic angular* is competitive and ranks second on LP-static, LP-broadcast, and EB. *Basic fitg* also leads on arc segments across all soccer datasets. APIDIS (basketball) is the only exception, where the simplest *basic parabola* wins, likely because short indoor trajectories do not benefit from drag or spin modeling.

Straight-segment performance varies little across models, so differences appear mainly on airborne segments. Detection noise, calibration error, and the inherent geometric ambiguity of monocular reconstruction, where multiple spin configurations yield nearly identical 2D projections, jointly limit how much complex models can exploit their additional parameters. This helps explain why *basic fitg*, whose free gravitational constant absorbs systematic bias into one effective parameter, outperforms *basic angular*: even on well-calibrated static cameras, angular-momentum parameters remain poorly constrained by 2D reprojection residuals from a single viewpoint.

The three MuJoCo variants perform comparably to ODE models of equivalent physical complexity on soccer datasets, indicating that the MuJoCo framework offers no clear advantage under noisy monocular observations. On APIDIS, they rank between parabola and fitg, suggesting that higher-fidelity fluid simulation may be better suited to controlled indoor settings with shorter camera-to-ball distances.

Table 5 breaks down LP results by camera type. Arc errors are substantially higher on the broadcast camera than on static cameras, while straight-segment errors increase

Table 5. Mean 3D error (m) by camera type on LP for the two best-performing models.

	LP-static			LP-broadcast		
Model	Full	Str	Arc	Full	Str	Arc
basic fitg	0.93	0.74	1.44	1.24	0.85	2.59
basic angular	0.89	0.67	1.48	1.20	0.80	2.62

only modestly. The gap is consistent with broadcast observation noise (pan-and-zoom calibration error, lower effective resolution, and motion blur) disproportionately affecting airborne segments, whose height recovery depends on accurate calibration far from the ground plane that anchors straight-segment fits.

6.3. Oracle 3D fitting

Table 4 reports oracle 3D fitting results. The *basic angular* model achieves the lowest error on four of five datasets and leads on arc segments across all five. The sole exception is SW, where *basic fitg* is marginally better on short sequences with limited flight diversity. This reverses the 2D ranking: accurate 3D positions resolve the geometric ambiguity that leaves angular-momentum parameters underdetermined from a single view. Fitting errors are also substantially lower on LP and SW (our multi-camera triangulation) than on EB and ISSIA-3D (external tracking), suggesting that ground-truth quality itself limits model evaluation.

6.4. Discussion and limitations

The ranking reversal between protocols reflects an observation-noise bottleneck: monocular geometric ambiguity prevents the spin-decomposition models from exploiting their extra parameters, so the single-parameter *basic fitg* model suffices under 2D fitting, whereas 3D inputs unlock the richer *basic angular* model. Our evaluation remains limited to free-flight segments and depends on reliable detections, calibration, pivot-point annotations, and dataset-specific ground-truth quality (Appendix 8.3 quantifies the resulting per-dataset coverage). More broadly, the lack of large-scale sports datasets with dense, accurate 3D ball annotations constrains evaluation scale and end-to-end automation.

7. Conclusion

We evaluated seven arc models under two complementary protocols. For soccer datasets, model rankings reverse between protocols: *basic fitg*, whose gravitational constant effectively captures unmodeled effects, leads on all soccer settings in monocular evaluation, yet *basic angular* with its spin decomposition wins on four of five datasets when fitted to 3D ground truth. This inversion shows that observation noise and monocular geometric ambiguity, not model expressiveness, are the current bottleneck. A same-match comparison on LP supports this interpretation: the broadcast view yields substantially higher arc error than static cameras, while straight-segment error rises only modestly, suggesting that broadcast observation quality especially limits airborne reconstruction accuracy where depth is most ambiguous. *Basic parabola* performs best on the basketball dataset (APIDIS), suggesting that complex models may offer limited benefit for sports without long outdoor ball trajectories.

Based on these results, we recommend matching model complexity to the observation regime: when only monocular video is available, improving detection and calibration is likely to yield larger gains than adding dynamical parameters, whereas multi-view or 3D inputs unlock the benefits of richer aerodynamic models.

To support further research, we release two multi-camera soccer datasets with triangulated 3D ground truth (Legia-Piast, 94 min, full match, and Stalowa Wola, 8.5 min, drills) plus a broadcast dataset with proprietary ground truth (aggregate results only), and segment-level annotations for APIDIS and ISSIA-3D.

Several directions remain open. First, uncertainty-aware fitting that propagates observation noise into model selection could automate the trade-off between *basic fitg* and *basic angular* on a per-segment basis. Second, joint trajectory–calibration optimization would address the dominant error source identified by our static-vs-broadcast comparison, potentially narrowing the monocular–oracle gap. Third, extending the framework beyond free-flight to model contacts, bounces, and player interactions would broaden coverage to full-match analysis.

Acknowledgements. This work was co-financed by the Polish National Centre for Research and Development (Narodowe Centrum Badań i Rozwoju – NCBiR) under Grant POIR.01.01.01-00-0718/21 as part of the project titled "Universal, automatic, artificial intelligence-based sports analytics system enabling acquisition & contextualization of 3D positional data from sports games and prediction of game events as well as assessment of tactics, skills and potential of teams & players".

References

[1] T. Bebie and Hanspeter Bieri. A video-based 3D-reconstruction of soccer games. *Computer Graphics Forum*, 2000. 2

[2] Mary Ann Branch, Thomas F. Coleman, and Yuying Li. A subspace, interior, and conjugate gradient method for large-scale bound-constrained minimization problems. *SIAM Journal on Scientific Computing*, 21(1):1–23, 1999. 4

[3] Marcello Davide Caio, Gabriel Van Zandycke, and Christophe De Vleeschouwer. Context-aware 3D object lo-

calization from single calibrated images: A study of basketballs. In *Proceedings of the 6th International Workshop on Multimedia Content Analysis in Sports*, page 49–54, 2023. 2

[4] Jordan Calandre, Renaud Péteri, Laurent Mascarilla, and Benoit Tremblais. Extraction and analysis of 3D kinematic parameters of table tennis ball from a single camera. In *2020 25th International Conference on Pattern Recognition (ICPR)*, pages 9468–9475, 2021. 2

[5] Hua-Tsung Chen, Ming-Chun Tien, Yi-Wen Chen, Wen-Jiin Tsai, and Suh-Yin Lee. Physics-based ball tracking and 3D trajectory reconstruction with applications to shooting location estimation in basketball video. *Soft Computing*, 20(3): 204–216, 2009. 2

[6] Damien Delannay, Nicolas Dallongeville, and Christophe de Vleeschouwer. Automatic sport video analysis in the APIDIS project. In *Proceedings of the ACM International Workshop on Video Surveillance and Sensor Networks (VSSN)*, pages 9–12, 2009. 3, 6

[7] Morten Holck Ertner, Sofus Schou Konglevoll, Magnus Ibh, and Stella Graßhof. SynthNet: Leveraging synthetic data for 3D trajectory estimation from monocular video. In *Proceedings of the 7th ACM International Workshop on Multimedia Content Analysis in Sports*, page 51–58. Association for Computing Machinery, 2024. 2

[8] Daniel Etaat, Dvij Kalaria, Nima Rahmanian, and Shankar Sastry. LATTE-MV: Learning to anticipate table tennis hits from monocular videos. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. 2

[9] Thomas Gossard, Andreas Ziegler, and Andreas Zell. TT3D: Table tennis 3D reconstruction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, 2025. 2, 3

[10] Marc Gutiérrez-Pérez and Antonio Agudo. SoccerNet-v3D: Leveraging sports broadcast replays for 3D scene understanding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, 2025. 2, 3, 6

[11] Chen Huang, Wen-Jiin Tsai, Suh-Yin Lee, and Jen-Yu Yu. Ball tracking and 3D trajectory approximation with applications to tactics analysis from single-camera volleyball sequences. *Multimedia Tools and Applications - MTA*, 60, 2012. 2

[12] Maxime Istasse, Vladimir Somers, Pratheeban Elancheliyan, Jaydeep De, and Davide Zambrano. DeepSportradar-v2: A multi-sport computer vision dataset for sport understandings. In *Proceedings of the 6th International Workshop on Multimedia Content Analysis in Sports*, page 23–29. Association for Computing Machinery, 2023. 3

[13] Daniel Kienzle, Julian Lorenz, Katja Ludwig, and Rainer Lienhart. Towards learning monocular 3D object localization from 2D labels using the physical laws of motion. *Proceedings of the International Conference on 3D Vision 2024 (3DV)*, 2024. 2

[14] Daniel Kienzle, Robin Schön, Rainer Lienhart, and Shin'Ichi Satoh. Towards ball spin and trajectory analysis in table tennis broadcast videos via physically grounded synthetic-to-real transfer. In *Proceedings of the IEEE/CVF International Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2025. 2, 3

[15] Hyunsung Kim, Hyun-Wook Choi, Chang Kim, Jinsung Yoon, and Sang-Ki Ko. Ball trajectory inference from multi-agent sports contexts using Set Transformer and hierarchical Bi-LSTM. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2023. 2

[16] Kenneth Levenberg. A method for the solution of certain non-linear problems in least squares. *Quarterly of Applied Mathematics*, 2(2):164–168, 1944. 4

[17] Paul Liu and Jui-Hsien Wang. MonoTrack: Shuttle trajectory reconstruction from monocular badminton video. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pages 3513–3522, 2022. 2, 3

[18] Donald W. Marquardt. An algorithm for least-squares estimation of nonlinear parameters. *Journal of the Society for Industrial and Applied Mathematics*, 11(2):431–441, 1963. 4

[19] William McNally, Jacob Lambeth, and Dustin Brekke. Combining physics and deep learning models to simulate the flight of a golf ball. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pages 5119–5128, 2023. 2

[20] J"urgen Metzler and Frank Pagel. 3D trajectory reconstruction of the soccer ball for single static camera systems. In *Proceedings of the IAPR Conference on Machine Vision Applications (MVA)*, pages 121–124, 2013. 2

[21] Timothy G. Myers and Stephen L. Mitchell. A mathematical analysis of the motion of an in-flight soccer ball. *Sports Engineering*, 16(1):29–41, 2013. 3

[22] Puntawat Ponglertnapakorn and Supasorn Suwajanakorn. Where is the ball: 3D ball trajectory estimation from 2D monocular tracking. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, 2025. 2, 3

[23] Gabriel Van Zandycke and Christophe De Vleeschouwer. 3D ball localization from a single calibrated image. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pages 3472–3480, 2022. 2

[24] Atsushi Yamada, Yoshiaki Shirai, and Jun Miura. Tracking players and a ball in video image sequence and estimating camera parameters for 3D interpretation of soccer games. In *Proceedings of the International Conference on Pattern Recognition (ICPR)*, pages 303–306, 2002. 2